



Survey Methods

2.2 - Target population, study population and sampling

Outline

- Definitions
- The concept of representativeness

Definitions

Population of inference

Population of interest: the whole 'universe' we are interested to study

Target population

Population we actually study



The estimates refer to this population

Sampling frame

Population from which we sample

Sample

Population selected for the survey

Respondents

Population from which data are collected

Definitions

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Sampling frame

Population from which we sample

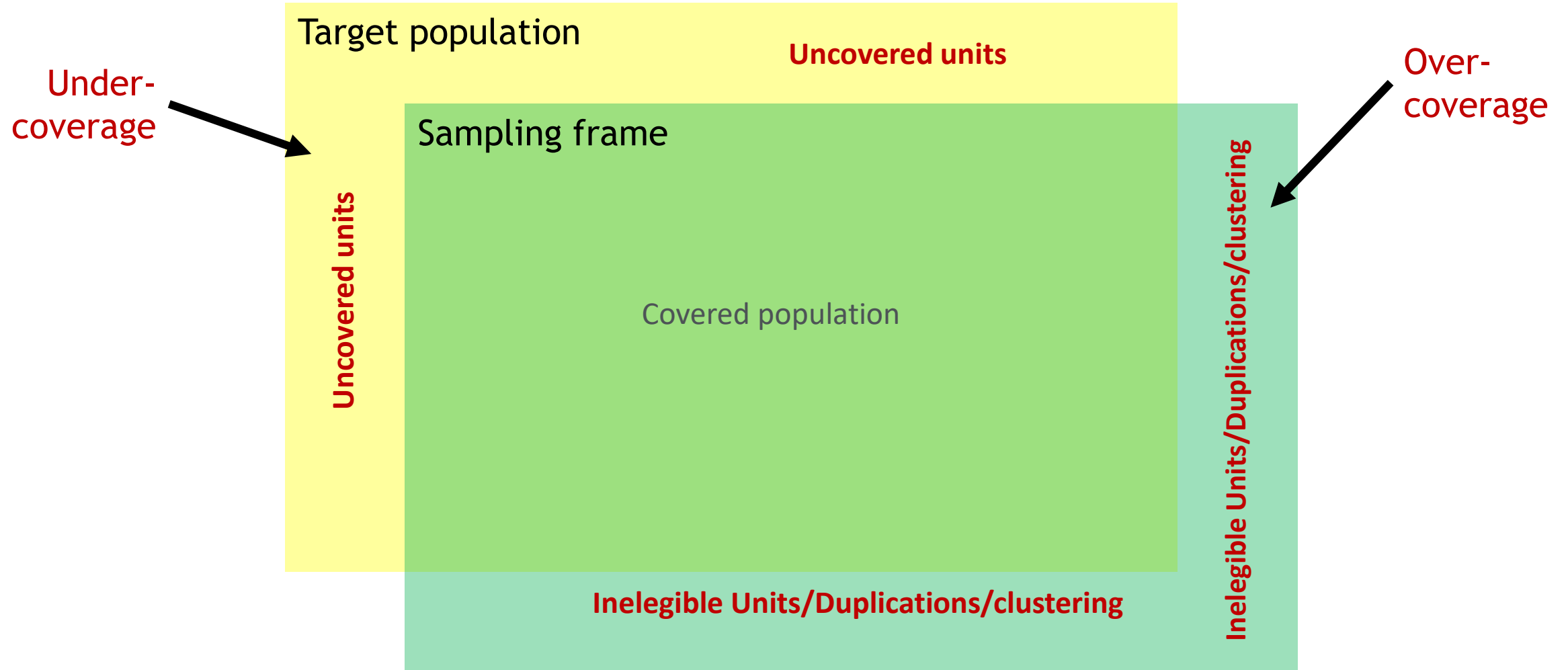
Sample

Population selected for the survey

Respondents

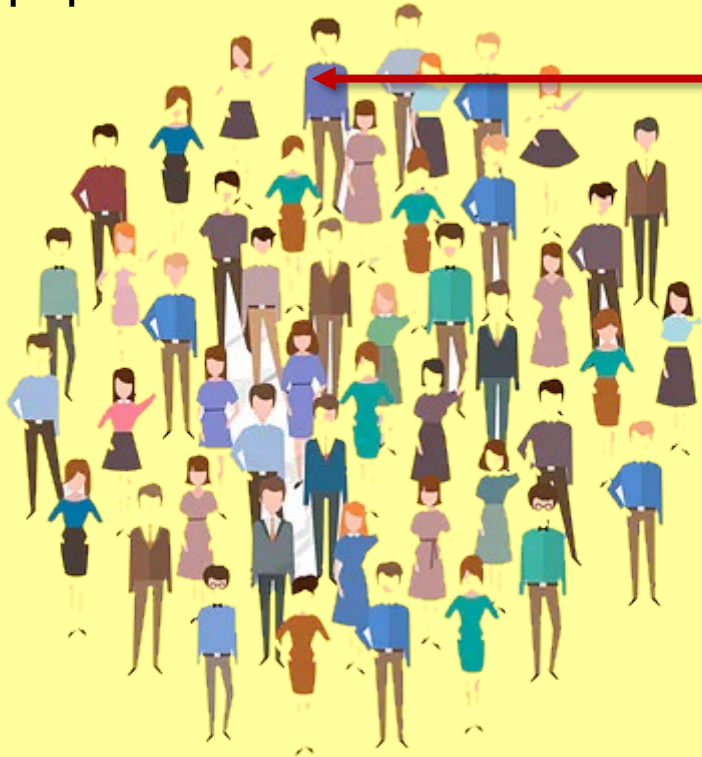
Population from which data are collected

Coverage and coverage error



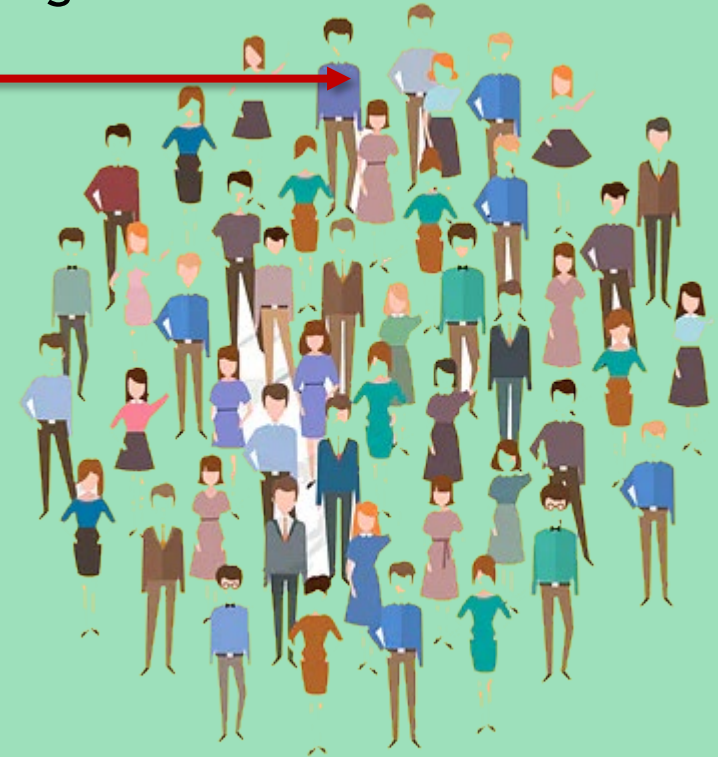
An ideal sampling frame?

Target population



1:1

Sampling frame



Examine the methodology section of the NIDS 2008 documentation, and extract this information:

- 1) Population of inference
- 2) Target population
- 3) Sampling frame
- 4) Sample
- 5) Respondents



05:00

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NIDS 2008

Population of inference ?

South African population

Target population ?

*'The target population for NIDS was private **households** in all nine provinces of South Africa and **residents in workers' hostels, convents and monasteries**. The frame excludes other collective living quarters such as students' hostels, old age homes, hospitals, prisons and military barracks'.*
(page 9)

Sampling frame ?

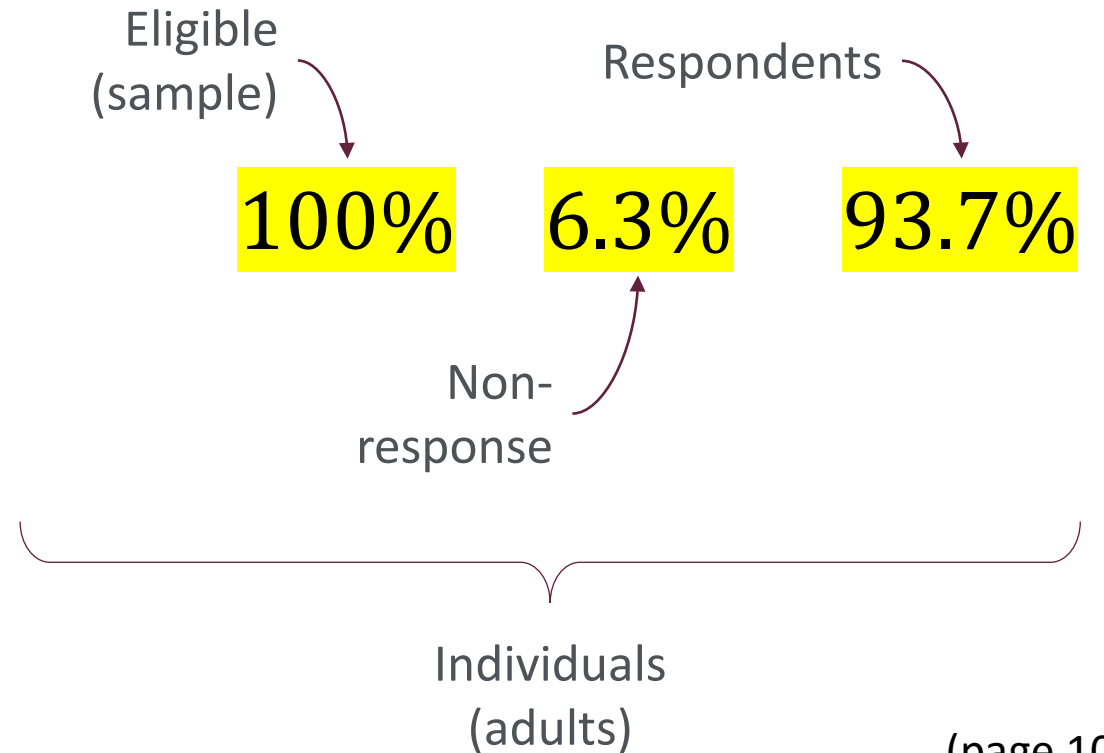
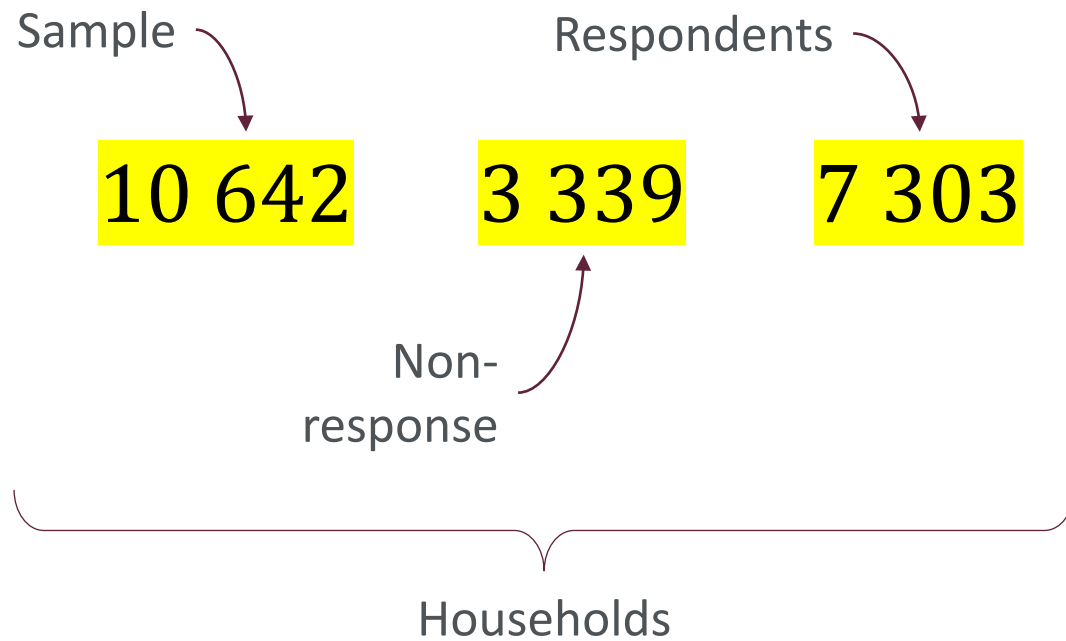
'...selected from Stats SA's 2003 Master Sample of 3000 PSUs. This Master Sample was the sample used by Stats SA for its Labour Force Surveys and General Household Surveys between 2004 and 2007 and for the 2005/06 Income and Expenditure Survey.'
(page 9)

Sample ?

all households in 400 psu randomly selected (multistage sampling, stratified by province) -> **10642 households**
(page 10+)

NIDS 2008

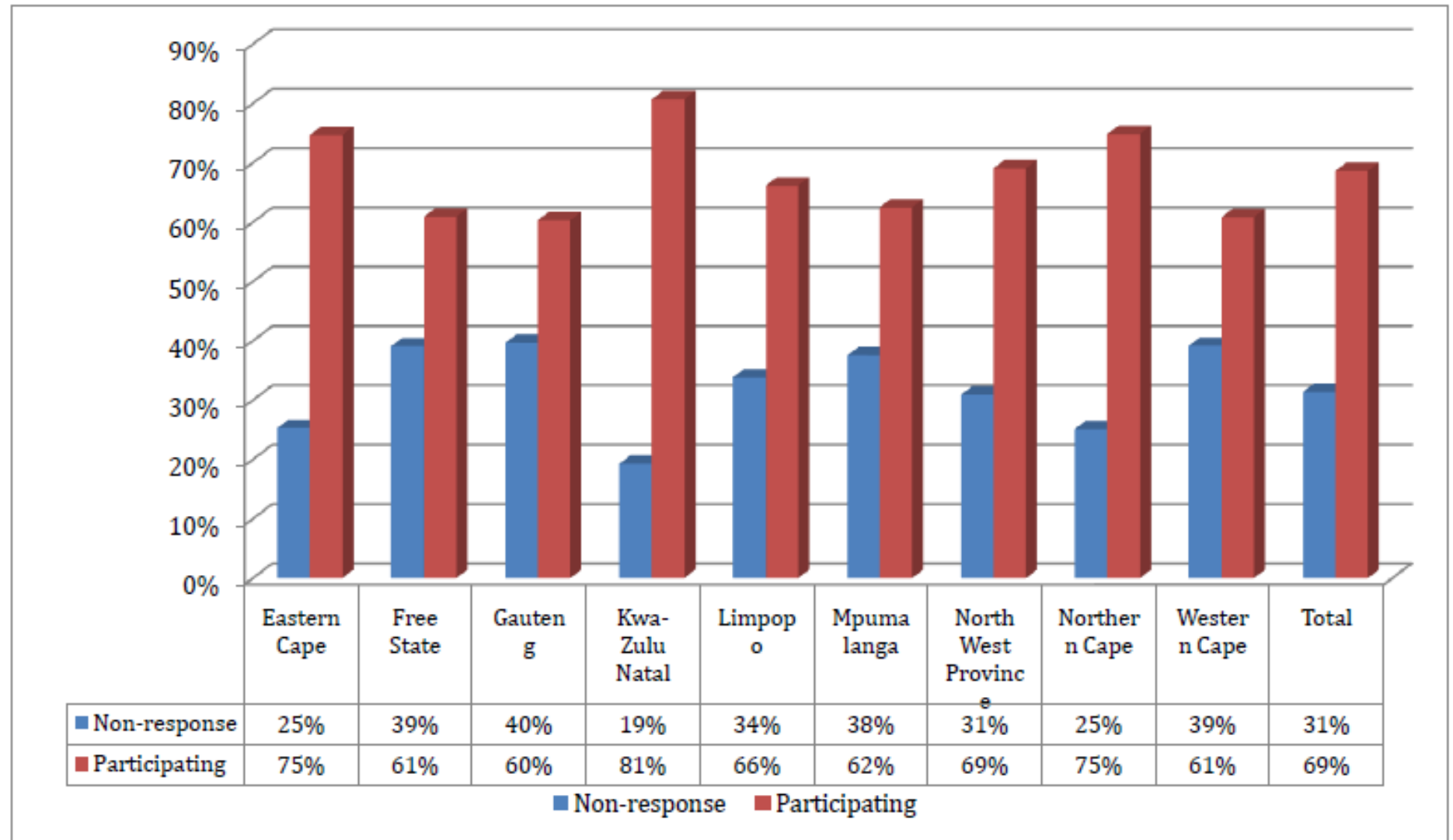
Respondents



(page 10+)

Random?

Figure 1: Response rates by province

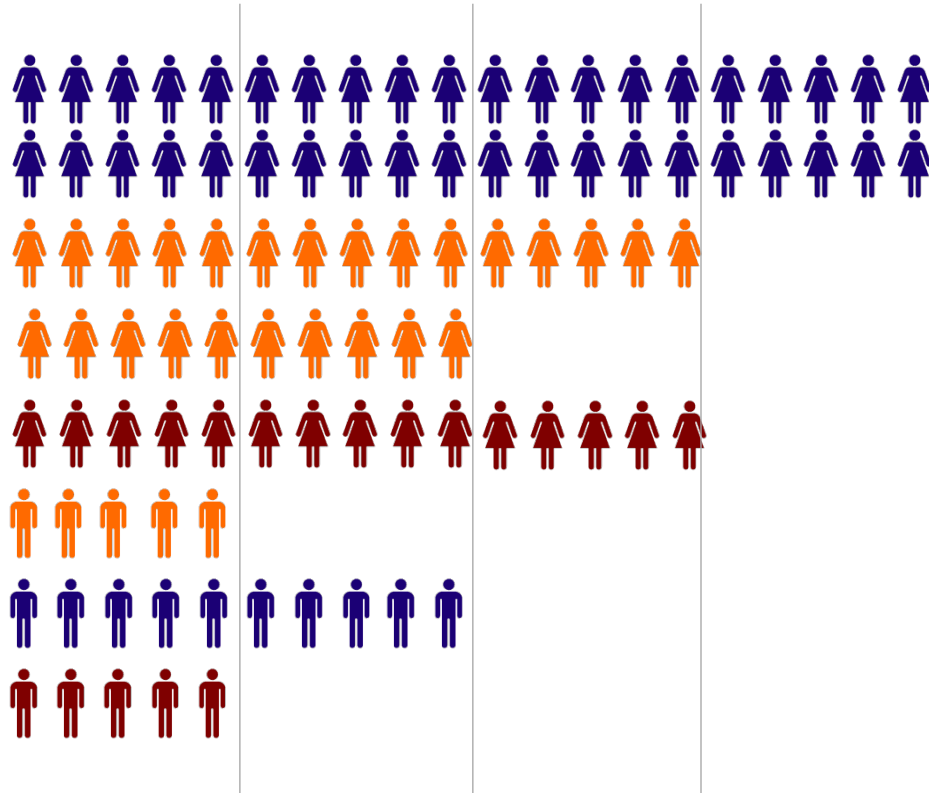


Representativeness



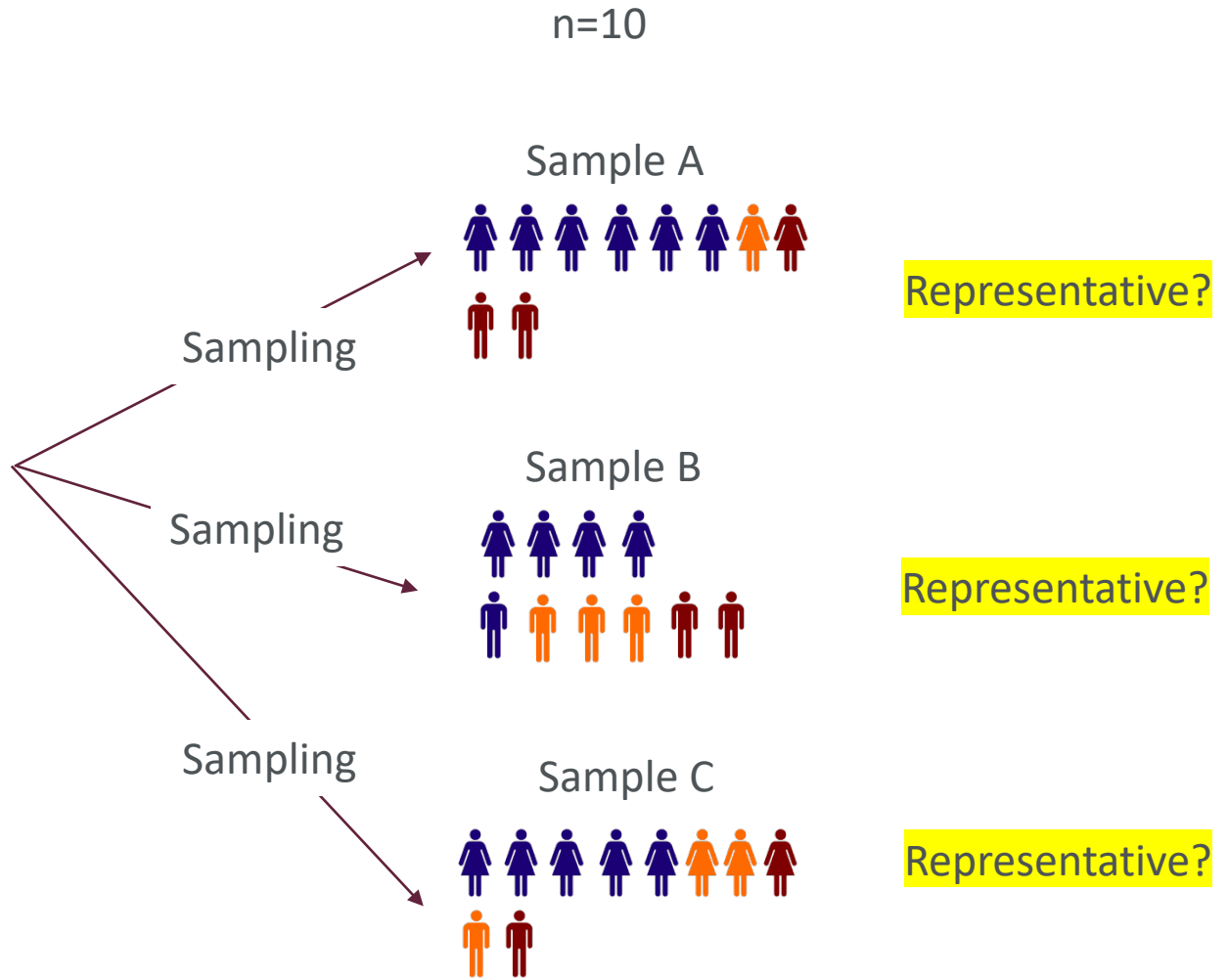
The sample is **similar** to the target population in **all characteristics of interest**

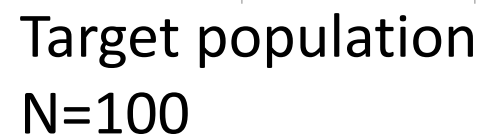




Target population
N=100

Characteristics of interest: Gender & hair colour



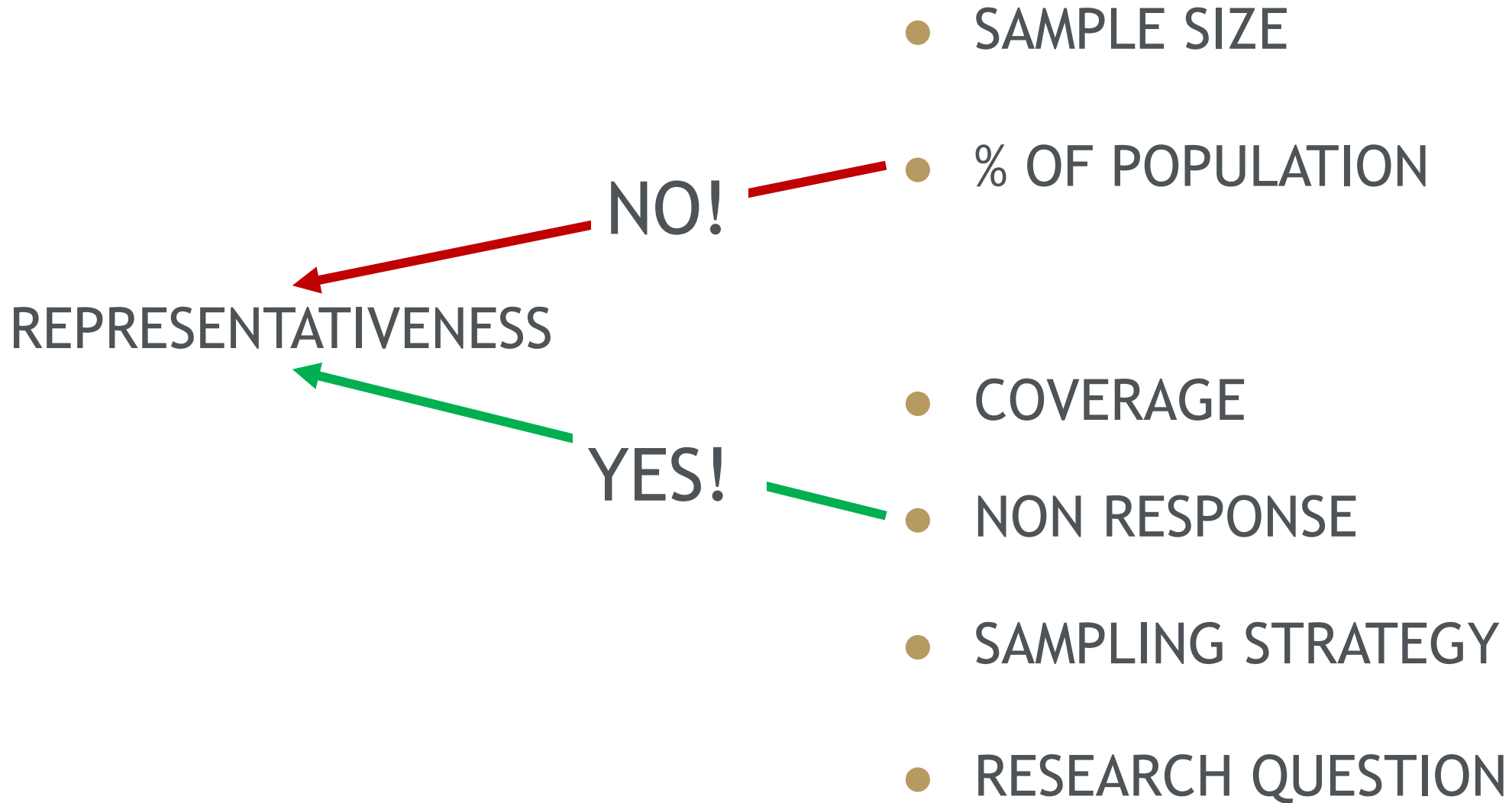


A diagram consisting of a single point on the left. From this point, two arrows branch out to the right. The upper arrow points to the word "Sampling", and the lower arrow points to the word "Sampling".

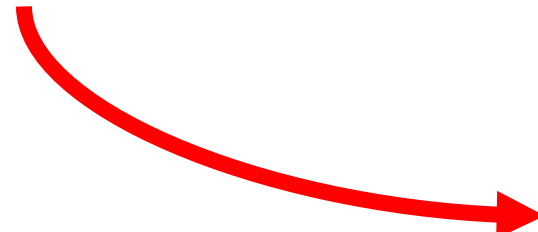


GENERALIZABILITY





How do we draw
representative samples
from a population?



Sampling
strategies

Summary questions

- Define population of inference, target population, sampling frame, sample, respondents
- What is coverage? And what is 'coverage error'?
- Define "representativeness" and "generalizability"

References

Groves, Robert M, Floyd J Fowler Jr, Mick P Couper, James M Lepkowski, Eleanor Singer, and Roger Tourangeau. 2009. *Survey Methodology*. John Wiley & Sons.

Leibbrandt, Murray, Ingrid Woolard, and Louise de Villiers. 2009. *Methodology: Report on NIDS Wave 1 . Technical Paper No. 1*. Southern Africa Labour and Development Research Unit. **Pages 9-12**.

Other resources

Cornesse, Carina. 2018. 'Representativeness and Response Quality of Survey Data'. Mannheim.
<https://madoc.bib.uni-mannheim.de/44066/>.



Thank you

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